

GOUGH ISLAND, Saint Helena

WMO: 689060

Lat: 40.3494S

Lon: 9.8803W

Elev: 54

StdP: 100.68

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	4.5	-1.9	3.2	5.6	-0.8	3.5	5.9	17.4	10.8	15.5	10.3	6.0	220	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.3	21.7	18.6	20.6	17.9	19.6	17.2	19.2	21.1	18.5	20.2	17.7	19.3	7.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.5	20.5	17.7	12.8	19.7	16.9	12.2	19.0	55.1	21.2	52.5	20.3	50.2	19.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	13.1	11.7	DB	1.9	24.0	1.4	1.2	0.9	24.8	0.1	25.5	-0.7	26.1	-1.7	27.0	(4)
(5)			WB	0.6	20.6	1.0	1.3	-0.2	21.6	-0.8	22.3	-1.3	23.1	-2.1	24.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	14.8	15.1	14.3	13.1	11.3	10.1	9.7	9.0	9.3	10.5	12.1	13.7	(6)	
	DBStd	3.22	2.43	2.57	2.45	2.48	2.49	2.41	2.63	2.30	2.36	2.29	2.40	2.38	(7)	
	HDD10.0	201	1	0	1	3	14	27	38	46	39	22	8	2	(8)	
	HDD18.3	2358	112	96	128	157	218	246	267	288	270	243	188	146	(9)	
	CDD10.0	896	150	142	133	97	55	31	29	16	20	37	70	116	(10)	
	CDD18.3	11	3	4	2	0	0	0	0	0	0	0	0	2	(11)	
	CDH23.3	6	1	4	1	0	0	0	0	0	0	0	0	1	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.8	5.6	5.6	5.4	5.4	5.9	6.2	6.2	6.3	6.1	5.6	5.7	5.6	(14)	
(15) Precipitation	PrecAvg	3185	217	191	252	272	299	323	274	311	276	287	238	250	(15)	
	PrecMax	3780	414	305	512	485	609	488	517	523	586	590	413	503	(16)	
	PrecMin	2576	76	66	93	136	121	147	135	130	61	114	42	108	(17)	
	PrecStd	334	76	54	87	77	95	87	79	100	92	90	82	87	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	23.8	22.0	19.9	18.5	16.9	17.0	15.7	15.9	17.8	19.7	22.4	(19)	
		MCWB	18.9	19.8	19.4	17.7	16.0	14.8	14.1	13.3	13.0	14.0	15.0	18.1	(20)	
	2%	DB	21.4	21.8	20.5	18.8	17.4	15.8	15.7	14.6	14.7	16.1	18.0	20.3	(21)	
		MCWB	18.2	19.0	18.3	16.5	15.4	14.0	13.6	12.7	12.5	13.4	15.1	17.1	(22)	
	5%	DB	20.1	20.7	19.5	18.0	16.4	15.0	14.7	13.7	13.9	15.0	16.9	18.8	(23)	
		MCWB	17.6	18.3	17.6	16.1	14.6	13.1	13.0	12.1	11.9	12.9	14.4	16.0	(24)	
	10%	DB	19.0	19.4	18.5	17.0	15.4	14.0	13.8	12.7	13.0	14.1	15.9	17.6	(25)	
		MCWB	16.6	17.1	16.6	15.3	13.8	12.3	12.2	11.1	11.3	12.2	13.6	15.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	20.7	19.8	18.1	17.0	15.5	14.9	14.0	14.0	15.1	16.7	19.2	(27)	
		MCDB	21.5	22.4	21.3	19.4	18.1	16.6	16.1	15.2	15.0	16.6	18.4	21.5	(28)	
	2%	WB	18.9	19.5	18.7	17.2	15.7	14.4	14.0	13.1	13.1	14.1	15.7	17.7	(29)	
		MCDB	20.8	21.4	20.0	18.4	16.8	15.6	15.1	14.3	14.2	15.5	17.4	19.5	(30)	
	5%	WB	18.0	18.6	17.9	16.4	14.8	13.5	13.2	12.3	12.4	13.3	14.9	16.6	(31)	
		MCDB	19.8	20.5	19.3	17.7	16.1	14.7	14.5	13.5	13.5	14.5	16.6	18.3	(32)	
	10%	WB	17.0	17.5	16.8	15.6	14.0	12.5	12.5	11.3	11.5	12.5	13.9	15.7	(33)	
		MCDB	18.7	19.3	18.2	16.9	15.4	13.8	13.8	12.6	12.8	13.9	15.5	17.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.3	4.6	4.2	4.0	3.9	3.8	4.0	4.0	4.4	4.7	5.1	(35)	
		MCDBR	6.4	6.6	6.0	5.0	4.9	4.6	4.6	5.0	5.0	5.3	6.0	6.4	(36)	
	5% WB	MCWBR	5.1	5.1	5.1	4.5	4.5	4.7	4.4	4.8	4.5	4.5	4.5	4.9	(37)	
		MCWBR	6.3	6.3	5.9	4.9	5.0	4.6	4.6	5.0	4.7	5.1	5.4	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.358	0.354	0.343	0.337	0.342	0.339	0.342	0.360	0.358	0.352	0.355	0.359	(40)	
		taud	2.530	2.554	2.581	2.587	2.507	2.470	2.476	2.417	2.445	2.488	2.508	2.513	(41)	
	Ebn at Noon		971	950	918	856	774	737	762	810	884	941	968	975	(42)	
		Edh at Noon	109	101	90	78	72	69	73	91	102	107	110	112	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.91	5.05	3.91	2.72	1.79	1.46	1.57	2.19	3.36	4.57	5.66	5.99	(44)	
	RadStd		0.33	0.29	0.24	0.19	0.12	0.09	0.08	0.10	0.15	0.27	0.38	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page